

Hongyi Pan, Ph.D.

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Experience

Postdoctoral Research Fellow

June 2023 – Present

Northwestern University, Advisor: Ulas Bagci

Chicago, IL, USA

- Develop and evaluate deep learning systems for medical image analysis spanning CNNs, Vision Transformers, multimodal models, foundation models, segmentation, classification, and domain generalization.
- Lead end-to-end development of **Cyst-X** and **LUMINA**, including dataset curation, preprocessing, model development, training, benchmarking, external validation, and clinical evaluation.
- Develop federated and domain-generalization methods using Fourier and wavelet transforms across heterogeneous breast, polyp, retinal, and pancreatic imaging datasets.
- Prototype and evaluate emerging vision and multimodal models, including **DINO-v2**, **Qwen2.5-VL**, and **Gemini 2.5 Pro**, for image and video understanding tasks.

Engineering Intern

May 2022 – Aug 2022

InnoPeak Technology (OPPO US)

Bellevue, WA, USA

- Developed attention-based variable-rate super-sampling models for real-time graphics rendering.
- Contributed to efficient deep learning models for real-time visual processing for cellphones.

Technical Skills

Deep Learning Models & Architectures: Vision Transformers (ViTs), MLLMs/VLMs (Qwen-VL, Gemini, LLaVA), Foundation Models (SAM, DINOv2), 3D CNNs (ResNet, DenseNet, MobileNet, 3D UNet), nnUNet, Diffusion Models, Graph Neural Networks (GNNs)...

Computer Vision & Algorithms: 3D/2D Medical Image Segmentation, Object Detection, Video Understanding & Temporal Reasoning, Domain Adaptation & Generalization, Frequency-Domain Attention Optimization...

Frameworks & Tools: PyTorch, MONAI, SimpleITK, OpenCV, Hugging Face, Scikit-learn, Docker, Linux/Ubuntu, Git, LaTeX...

Selected Projects

Cyst-X: Federated AI for Pancreatic Cancer Risk Prediction [Cyst-X](#) [🔗](#)

- Built an end-to-end AI pipeline for pancreatic MRI analysis, combining pancreas segmentation (**PanSegNet**), 3D DenseNet-121 classification, and federated learning across 7 clinical centers.
- Led a publicly available dataset project comprising **1,461 MRI scans from 764 patients**; evaluated model performance against radiologists and clinical guidelines.

Multi-Vendor Mammography and Domain Generalization *CVPR 2026; MICCAI Deep-Brea3th Workshop 2026 – Best Paper Award* [LUMINA](#) [🔗](#) | [BreastMammo/DenseMammo](#) [🔗](#)

- Curated and benchmarked multi-view FFDM datasets, including **LUMINA** (1,824 images from 468 patients across 6 scanner vendors) and **BreastMammo/DenseMammo** (3,374 images from 1,067 patients) with pathology, BI-RADS, density, and energy metadata.
- Developed a training-free, foreground-only pixel-space histogram-matching protocol and extended it to a target-free domain-generalization framework with inter-view style synchronization to reduce vendor- and energy-induced domain shifts.
- Benchmarked multi-view CNN and Vision Transformer architectures across three clinical tasks, achieving **93.54% AUC** for pathology-confirmed diagnosis and outperforming MixStyle and DFT on unseen external cohorts.

VideoAds: Multimodal Large Language Models for Video Understanding *ICCV 2025* [VideoAds](#) [🔗](#)

- Developed a video QA benchmark for evaluating **multimodal large language models** on fast-paced videos with dense temporal events.
- Developed a quantitative video-complexity metric using **DINO-v2** feature variance to characterize temporal coherence, rapid scene transitions, and long-range semantic progression.
- Benchmarked leading multimodal models, including **Qwen2.5-VL** and **Gemini 2.5 Pro**, on visual summarization and causal inference tasks.

Efficient Deep Learning and Vision Transformers *ICML 2023, TNNLS 2024, IJCAI 2026*

- Developed efficient neural network architectures using **DCT**, **wavelet**, and **Hadamard transforms** across

CNNs and Vision Transformers.

- Designed structure-preserving initialization and frequency-domain compression strategies for Vision Transformers.
- Developed and evaluated efficient CNN layers and a hybrid quantum-classical Hadamard layer on datasets including ImageNet-1K and COCO.

Federated Domain Generalization for Medical Image Analysis *ICASSP 2024, ICASSP 2025*

- Developed DFT-based federated domain-generalization methods for heterogeneous medical imaging data.
- Evaluated the methods across breast, polyp, and retinal fundus datasets.

Wildfire Detection on Edge Devices *TECS 2022*

- Developed a **MobileNetV3**-based real-time wildfire detection model optimized for edge deployment.
- Deployed the system on an **NVIDIA Jetson Nano** using HPWREN surveillance data.

Grants

Cyst-Suite: A Federated Foundational Model for Pancreatic Cyst Management *Aug 2026 – Jul 2028*

- NIH K99/R00 Pathway to Independence Award (1K99EB039097-01). **Role: PI. \$212,582** K99 phase.

Education

University of Illinois Chicago

Aug 2019 – Aug 2023

Ph.D. in Electrical and Computer Engineering (Advisor: Ahmet Enis Cetin)

Chicago, IL, USA

Dissertation: *Orthogonal Transform Domain Based Neural Network Layers*. Focus on efficient neural networks, anomaly detection, and embedded edge vision.

University of Illinois Chicago

Aug 2017 – May 2019

M.S. in Electrical and Computer Engineering

Chicago, IL, USA

Chang'an University

Sep 2014 – Jul 2018

B.S. in Automation

Xi'an, Shaanxi, China

Selected Publications

Published 31 peer-reviewed conferences and 23 journals. Google Scholar citation: 100+; h-index: 19; i10-index: 27.

Full publication list: <https://www.scholar.google.com/citations?user=23I1CMYAAAAJ>

H. Pan, et al. BreastMammo and DenseMammo: Benchmarks for Mammography Domain Generalization. *Medical Image Computing and Computer Assisted Intervention (MICCAI) Workshop Deep-Brea3th, 2026*. **Best paper award**.

H. Pan, et al. Discrete Cosine Transform Based Decorrelated Attention for Vision Transformers. *International Joint Conference on Artificial Intelligence (IJCAI)*, 2026.

H. Pan, et al. LUMINA: A Multi-Vendor Mammography Benchmark with Energy Harmonization Protocol. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

H. Pan, et al. Cyst-X: A Federated AI System Outperforms Clinical Guidelines to Detect Pancreatic Cancer Precursors and Reduce Unnecessary Surgery. *Nature Communications*, under review, major revision.

H. Pan, et al. Frequency-Based Federated Domain Generalization for Polyp Segmentation. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.

Z. Zhang, M. Dou, L. Peng, **H. Pan**, U. Bagci, B. Gong. VideoAds for Fast-Paced Video Understanding: Where Opensource Foundation Models Beat GPT-4o & Gemini-1.5 Pro. *International Conference on Computer Vision (ICCV)*, 2025.

H. Pan, et al. Multichannel Orthogonal Transform-Based Perceptron Layers for Efficient ResNets. *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, 2024.

H. Pan, et al. Domain Generalization with Fourier Transform and Soft Thresholding. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

B. Wang, **H. Pan**, et al. GazeGNN: A Gaze-Guided Graph Neural Network for Chest X-Ray Classification. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.

H. Pan, et al. A hybrid quantum-classical approach based on the Hadamard transform for the convolutional layer. *International Conference on Machine Learning (ICML)*, 2023.

H. Pan, et al. Real-time wireless ecg-derived respiration rate estimation using an autoencoder with a dct layer. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

M. Yang, Z. Ye, **H. Pan**, M. Farhat, A.E. Cetin, P.Y. Chen. Electromagnetically unclonable functions generated by non-Hermitian absorber-emitter. *Science Advances*, 2023.

H. Pan, et al. Block walsh-hadamard transform-based binary layers in deep neural networks. *ACM Transactions on Embedded Computing Systems (TECS)*, 2022